



UNIVERSITY OF
LIVERPOOL

SECOND SEMESTER EXAMINATIONS 2024/25

CYBERSECURITY

TIME ALLOWED : Two Hours

INSTRUCTIONS TO CANDIDATES

Answer FOUR out of FIVE questions.

All questions have the same maximum mark of 25.

If you answer five questions, the question with the lowest mark will be disregarded.

Calculators are NOT permitted.

Answer **FOUR** out of **FIVE** questions.

1. (a) Describe the packet-filtering technique used in firewalls. What are advantages and disadvantages of this technique?
(6 marks)
(b) What is a weak collision resistance property of a hash function and what the role does it play in applications of hash functions?
(6 marks)
(c) Propose a variant of the Diffie-Hellman (DH) Key Exchange algorithm which would allow to exchange secret keys between *three* participants. How do you argue it is secure algorithm?
(13 marks)
2. (a) How is protection against passive attacks guaranteed in BB84 quantum key exchange algorithm?
(10 marks)
(b) Explain why RSA can be considered as a partially homomorphic encryption algorithm.
(7 marks)
(c) List three main categories of authentication techniques. Discuss advantages and disadvantages of using physical or behavioural characteristics for authentication.
(8 marks)
3. (a) Describe how a digital signature can be realised using a public-key cryptosystem.
(6 marks)
(b) Describe briefly 4 levels of information protection and the methods used at each level.
(4 marks)
(c) What is a mix-network? Describe functioning of a mix node and the process of communication in a mix-network.
(7 marks)
(d) Explain what are Secure Multi Party Computations and give an example of their possible applications.
(8 marks)
4. (a) What is an iteration count in implementations of password-based encryption? How may setting different values to iteration count affect the brute force search attack against an encryption algorithm?
(7 marks)
(b) What are the two main categories of intrusion detection methods? Describe advantages and disadvantages of methods from each category.
(5 marks)
(c) A simple key exchange protocol shown below can be used to establish a key over an open channel. Here K_a is a public key of A, N_a and N_b are random numbers (nonces) chosen by A and B, respectively, and k_{ab} is the key for symmetric encryption to be exchanged.

- Message 1. $A \rightarrow B : N_a$
- Message 2. $B \rightarrow A : N_b$
- Message 3. $A \rightarrow B : \{k_{ab}, A, N_b\}_{K_a}$
- Message 4. $B \rightarrow A : \{N_a\}_{k_{ab}}$

What is the possible issue with this protocol? How can it be fixed?

(9 marks)

(d) What is aggregated information? What is anonymized information? Is privacy protected if only some records of anonymized and aggregated information are known about a person?

(4 marks)

5. (a) For a block cipher, what is the Electronic Codebook mode (ECB)? Describe the main disadvantage of this mode and how it can be fixed.

(7 marks)

(b) What is the reason for the anonymity protection system *Tor* to use both symmetric and public-key encryption?

(8 marks)

(c) What is a purpose of using multiple layers of encryption in CryptDB approach to querying encrypted databases?

(10 marks)